

1. Cover Page

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Program/Batch: B083 - Networkwalks

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Modules completed: W3-PM1, W3-PM2

2. Liability Disclaimer — Copy from your Week 2 report.

3. Introduction

"This report covers password cracking using two methods: John the Ripper (JTR) on Kali Linux (W3-PM1) and Networkwalks web tools (W3-PM2). Both modules demonstrate how weak passwords can be recovered and why strong password policies are essential."

4. Tools Used

Tool	Purpose
Kali Linux	Operating system for JTR
pdf2john	Extract password hash from PDF
John the Ripper (JTR)	Crack password hash
Networkwalks Hash Calculator	Extract hash from locked PDF (web-based)
Networkwalks Password Cracker	Crack hash (web-based)

5. Activities Performed

5.1 W3-PM1: Password Cracking with JTR

"I downloaded the encrypted PDF file (My_Locked_PDF3.pdf) and used pdf2john to extract the hash. The hash started with \$pdf\$ as expected. I saved the hash to hash3.txt and ran John the Ripper. John successfully cracked the password as 1qaz2wsx within seconds. I verified by opening the PDF with the cracked password."

5.2 W3-PM2: Password Cracking with Networkwalks Tools

"I used the Networkwalks Hash Calculator to extract the hash from My_Locked_PDF2.pdf via the web browser. I then pasted the hash into the Networkwalks Password Cracker, which successfully recovered the password password1. This demonstrated that password cracking can be performed entirely through web-based tools without any local installation."

6. Risk Analysis

#	Finding	Impact	Risk Level
1	Weak password (1qaz2wsx) on PDF3 wordlists	Easily cracked in seconds using common wordlists	High
2	Weak password (password1) on PDF2	Cracked instantly with dictionary attack	High
3	PDF encryption easily bypassed	Confidential data exposed	High

7. Recommendations

Use strong, unique passwords (12+ characters, mixed case, numbers, symbols).

Avoid dictionary words and common patterns.

Use password managers to generate and store strong passwords.

Enable file-level encryption with strong passphrases.

8. Conclusion

"In Week 3, I learned how password cracking works using both command-line (JTR) and web-based tools (Networkwalks). The exercise demonstrated that weak passwords can be cracked in seconds, emphasizing the importance of strong password policies. I also learned how password hashes are extracted from encrypted files and how dictionary attacks work."

9. Evidence

```
kali-linux-2026.1-virtualbox-amd64 (Clean Kali - Week 1) [Running] - Oracle VirtualBox : 1
File Machine View Input Devices Help
1 2 3 4
EN 13:36

kali@kali: ~
Session Actions Edit View Help
└─$ pdf2john My_Locked_PDF3.pdf > hash3.txt
File not found: My_Locked_PDF3.pdf

(kali@kali)-[~]
└─$ la
.bash_logout      .local            .vboxclient-clipboard-tty7-service.pid
.bashrc           .maltego         .vboxclient-draganddrop-tty7-control.pid
.bashrc.original .mozilla         .vboxclient-draganddrop-tty7-service.pid
.BurpSuite       .msf4            .vboxclient-hostversion-tty7-control.pid
.cache           Music            .vboxclient-seamless-tty7-control.pid
.config          'My Locked PDF3.pdf'
.Desktop         .Pictures        .vboxclient-seamless-tty7-service.pid
.dmrcc           .profile         .vboxclient-vmvga-session-tty7-control.pid
.Documents       Projects         .vboxclient-vmvga-session-tty7-service.pid
.Downloads       protonvpn-stable-release_1.0.8_all.deb
.face            Public
.face.icon       .recon-ng
.gnupg           scam_analysis.txt
hash3.txt        scam_source.txt
.ICEauthority    .ssh
index.html       .sudo_as_admin_successful
.ipython         Templates
.java           .vboxclient-clipboard-tty7-control.pid
               .Videos
               .wget-hsts
               wrk
               .Xauthority
               .xsession-errors
               .xsession-errors.old
               .zenmap
               .zprofile
               .zsh_history
               .zshrc

(kali@kali)-[~]
└─$ pdf2john My Locked PDF3.pdf > hash3.txt
File not found: My
File not found: Locked
File not found: PDF3.pdf

(kali@kali)-[~]
└─$ pdf2john My_Locked_PDF3.pdf > hash3.txt
File not found: My_Locked_PDF3.pdf

(kali@kali)-[~]
└─$ pdf2john MyLockedPDF3.pdf > hash3.txt

(kali@kali)-[~]
└─$ cat hash3.txt
MyLockedPDF3.pdf:$pdf$4*4*128*-1028*1*16*34eb542eff4e1b0b32d25ce15a9a7281*32*b77872bfc9a24fb2f845066283a8fc1b0021446990b9e41140
256e4b552cd57988f5134214b91920d94d7a6bf550ea94a2995c7f2ab02

(kali@kali)-[~]
└─$
```

```
kali-linux-2026.1-virtualbox-amd64 (Clean Kali - Week 1) [Running] - Oracle VirtualBox : 1
File Machine View Input Devices Help
1 2 3 4
EN 13:36

kali@kali: ~
Session Actions Edit View Help
.recon-ng
.scam_analysis.txt
.scam_source.txt
.ssh
.sudo_as_admin_successful
Templates
.vboxclient-clipboard-tty7-control.pid
.Xauthority
.xsession-errors
.xsession-errors.old
.zenmap
.zprofile
.zsh_history
.zshrc

(kali@kali)-[~]
$ pdf2john My_Locked_PDF3.pdf > hash3.txt
File not found: My
File not found: Locked
File not found: PDF3.pdf

(kali@kali)-[~]
$ pdf2john My_Locked_PDF3.pdf > hash3.txt
File not found: My_Locked_PDF3.pdf

(kali@kali)-[~]
$ pdf2john MyLockedPDF3.pdf > hash3.txt

(kali@kali)-[~]
$ cat hash3.txt
MyLockedPDF3.pdf:$pdf$4*4*128*-1028*1*16*34eb542eff4e1b0b32d25ce15a9a7281*32*b77872bfc9a24fb2f845066283a8fc1b0021446990b9e41140
256e4b552cd57988f5134214b91920d94d7a6bf550ea94a2995c7f2ab02

(kali@kali)-[~]
$ john hash3.txt
Created directory: /home/kali/.john
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 7 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
1qaz2wsx (MyLockedPDF3.pdf)
1g 0:00:00:00 DONE 2/3 (2026-09-21 13:36) 5.263g/s 81815p/s 81815c/s 81815C/s pretty..hermosa
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~]
$
```

kali-linux-2026.1-virtualbox-amd64 (Clean Kali - Week 1) [Running] - Oracle VirtualBox : 1

File Machine View Input Devices Help

EN 13:36

kali@kali: ~

Session Actions Edit View Help

.java .vboxclient-clipboard-tty7-control.pid .zshrc

```
(kali@kali)-[~]
$ pdf2john My_Locked_PDF3.pdf > hash3.txt
File not found: My
File not found: Locked
File not found: PDF3.pdf

(kali@kali)-[~]
$ pdf2john My_Locked_PDF3.pdf > hash3.txt
File not found: My_Locked_PDF3.pdf

(kali@kali)-[~]
$ pdf2john MyLockedPDF3.pdf > hash3.txt

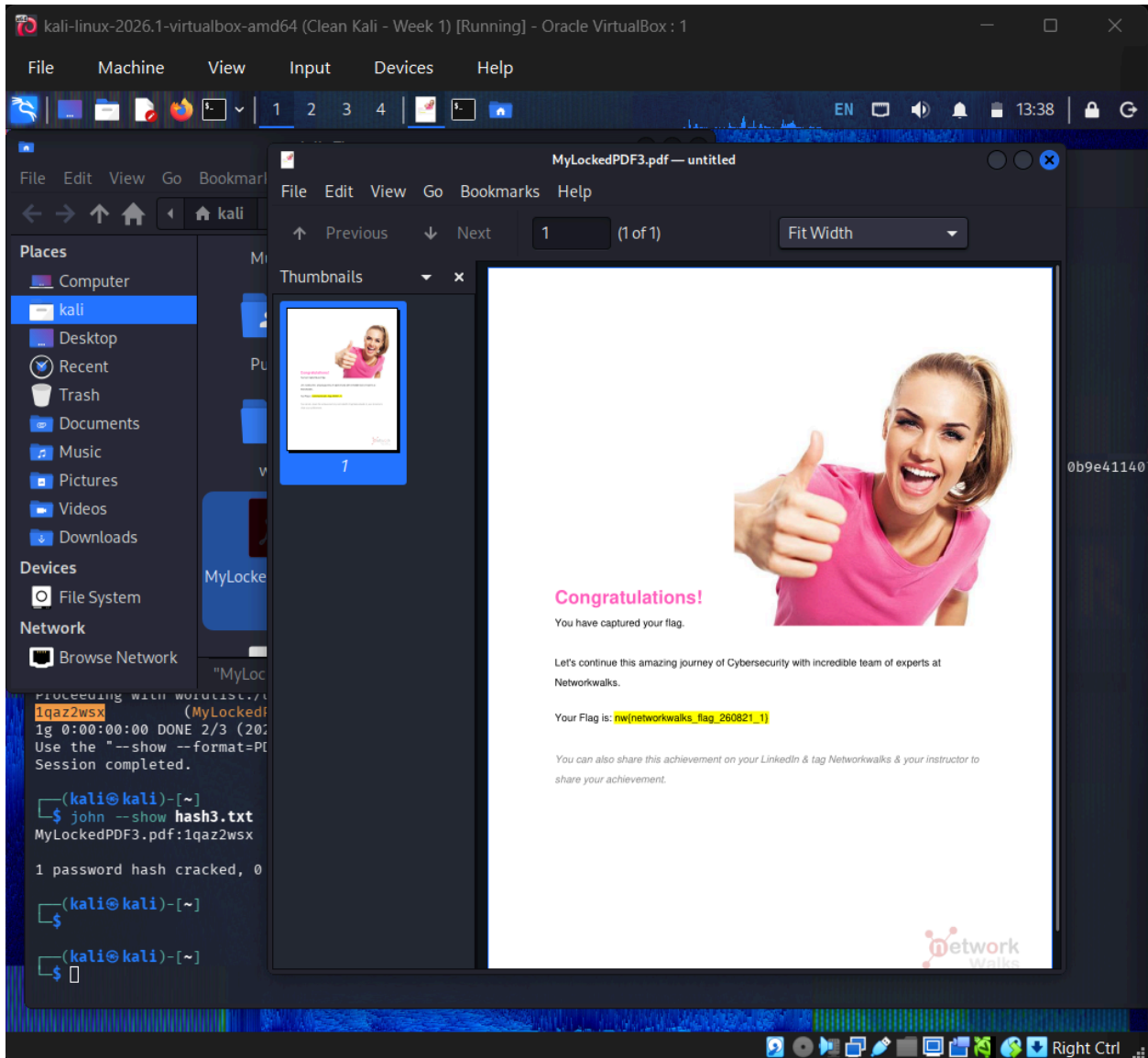
(kali@kali)-[~]
$ cat hash3.txt
MyLockedPDF3.pdf:$pdf$4*4*128*-1028*1*16*34eb542eff4e1b0b32d25ce15a9a7281*32*b77872bfc9a24fb2f845066283a8fc1b0021446990b9e41140
256e4b552cd57988f5134214b91920d94d7a6bf550ea94a2995c7f2ab02

(kali@kali)-[~]
$ john hash3.txt
Created directory: /home/kali/.john
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 7 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
1qaz2wsx (MyLockedPDF3.pdf)
1g 0:00:00:00 DONE 2/3 (2026-09-21 13:36) 5.263g/s 81815p/s 81815c/s 81815C/s pretty..hermosa
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~]
$ john --show hash3.txt
MyLockedPDF3.pdf:1qaz2wsx

1 password hash cracked, 0 left

(kali@kali)-[~]
$
```



Hash Calculator - Networkwalks

https://networkwalks.com/hash-calculator/

Chat

Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF

TestFilePDF

Click to choose a PDF, or drop one here

Nothing is uploaded, the PDF is parked locally

My Locked PDF2.pdf | 209.7 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

5a0f8454*1208*582371516*9003f2cdeb1f05c0ff09d226d0b1eaf12*9f13c8a39ad9743
64d16a077d767000214469080e4114071a40104984c1*32*cecdc74b1955ae2885d3b352
4e1374c95cbb0cc3c45316494d9446ef81af1

CopyDownload

Revision: 04 Version: 04 Key length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

Free tool by Networkwalks Academy | Cybersecurity & Ethical Hacking with AI

We Are Here!

